

HTB Sherlock: Fruitzzy

Write-Up • DFIR / Digital Forensics

Platform: Hack The Box | Solved by: RismaFareedh | Sherlock Rank: #31 | Solve Date: 01 May 2026

Scenario

CyberJunkie started out as a junior QA Analyst at his friend's startup. He called the CEO of the startup because he believed he had mistakenly downloaded something malicious. The CEO sought help from you, his friend in the cybersecurity field. You sent him a guide on collecting evidence from the machine using KAPE. Now you have been given the forensic image so you can analyze and help your friends, as they cannot afford to hire an MSSP.

A classic phishing-to-RMM-backdoor chain. We are handed a KAPE-collected forensic image and tasked with reconstructing the full attack from the initial phishing email all the way to persistent remote access.

Tools Used

- PhishTool: email header analysis
- DB Browser for SQLite: browser history / downloaded artifact analysis
- Eric Zimmerman's Registry Explorer: Amcache.hve parsing
- Eric Zimmerman's MFTECmd: Master File Table analysis (parsed to CSV, reviewed in Excel)
- Windows Event Viewer: Windows Defender & Service Control Manager event logs
- VirusTotal: file hash and domain threat intelligence, domain registration lookup

Executive Summary

Incident Type	Phishing leading to Remote Monitoring and Management (RMM) Backdoor Deployment
Target Environment	Windows Endpoint (QA-Pipeline-02)
Threat Actor Vector	Social engineering via a themed email lure ("Special Party Invitation") directing the user to a malicious payload download link hosted on pomi.digital
Impact	Full, persistent endpoint compromise via unauthorized installation of Datto RMM (CentraStage), granting the attacker administrative remote access with LocalSystem privileges

Task 1 — What is the Subject/topic of the Phishing email?

Investigation

The first step was uploading the suspicious .eml file to PhishTool for header analysis. The tool immediately surfaced the full email metadata. The email carried the subject **"Special Party Invitation from JANET CARNAHAN"**, sent from **businessinvitation432@gmail.com**, Display Name "Business invitation", to **CyberJunkieContractor@gmail.com**, and Timestamp **"2026-03-04T14:48:56Z"**. The originating IP was **209.85.220.65**, which reverse-resolved to **mail-sor-f65.google.com**. The email body was deliberately brief: *"Please find your exclusive invitation at the following link. We look forward to welcoming you at the event. Regards, Janet."* — a social-engineering lure designed to drive a single click.

PhishTool

DashboardUploadsIn-trayNotifications▼My Account▼

Uploads > Special Party Invitation from JANET CARNAHAN

Special Party Invitation from JANET CARNAHAN

DetailsAuthenticationURLsAttachmentsTransmissionX-headers

RenderedHTMLPlaintextSource

From

businessinvitation432@gmail.com

...

Display name

Business invitation

Sender

None

To

CyberJunkieContractor@gmail.com

Cc

None

In-Reply-To

None

Timestamp

2026-03-04T14:48:56Z

Reply-To

None

Message-ID

<CADM-727bnHfFdux484CZah4CFRdriqHY8S=j90bQQTfUtrMGgw@mail.gmail.com>

Return-Path

businessinvitation432@gmail.com

...

Originating IP

209.85.220.65 (Received-SPF) ▼

...

rDNS

mail-sor-f65.google.com

Please find your exclusive invitation at the following link

We look forward to welcoming you at the event.

Regards

Janet

Details

Authentication

URLs

Attachments

Transmission

X-headers

Rendered

HTML

Pla

Filters (0) ▼

URL

Domain

VirusTotal

https://pomi.digital/premium?__cf_chl_tk=6SX3C1utT5mbwmlZyMDU.WICiyPkGsBkaOTitVBss-1772039836-1.0.1.1-JvMwxVsDmAF0zIVuaF0XNh73a8usyJzPeFDOJpaT4j8

pomi.digital

Configure

Please find your exclusi

We look forward to wel

Regards

Janet

Task 2 — What is the malicious URI that the malicious link redirected to?

Investigation

To trace where the phishing link led, I examined the Microsoft Edge browser history using DB Browser for SQLite, opening the History database at:

H:\C\Users\cyberjunkie\AppData\Local\Microsoft\Edge\User Data\Default\History

Browsing the downloads table revealed a single download entry. The tab_url field contained the full malicious URI:

https://pomi.digital/premium/windows_download.php

DB Browser for SQLite - H:\C\Users\cyberjunkie\AppData\Local\Microsoft\Edge\User Data\Default\History

File

Edit

View

Tools

Help

New Database

Open Database

Write Changes

Revert Changes

Undo

Open Project

Save Project

Attach Database

Close Database

Database Structure

Execute SQL

Browse Data

Edit Pragma

Table: downloads

Filter in any column

	tab_url	tab_referrer_url	http_method	by_ext_j
1	https://pomi.digital/premium/windows_download.php	https://pomi.digital/premium?...		

Edit Database Cell

Mode: Text

1 https://pomi.digital/premium/windows_download.php

Task 3 — What is the name of the downloaded file?

Investigation

This was corroborated by the Master File Table (MFT). Using MFTECmd, the MFT was parsed to a CSV and opened in Excel. Filtering on the **ParentPath** as 'Downloads' returned row 226031 with the following attributes:

- ParentPath: .\Users\cyberjunkie\Downloads
- FileName: premium.exe

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
1	EntryNu	Sequen	InUse	Parent	Parent	ParentPath	FileName	Extensi	FileSize	Referer	Repars	IsDirec	HasAds	IsAds	SI<FN	uSecZe	Copied
50609	44433	1	TRUE	18334	1	.\Windows\WinSxS\amd64_micros	desktop.ini	.ini	174	1		FALSE	FALSE	FALSE	TRUE	FALSE	FALSE
50610	44434	1	TRUE	3586	1	.\Users\Public\Downloads	desktop.ini	.ini	174	1		FALSE	FALSE	FALSE	TRUE	FALSE	TRUE
172989	98742	2	TRUE	95037	4	.\Users\cyberjunkie\Downloads	desktop.ini	.ini	282	1		FALSE	FALSE	FALSE	FALSE	FALSE	FALSE
226031	149642	9	TRUE	95037	4	.\Users\cyberjunkie\Downloads	premium.exe	.exe	11056040	1		FALSE	FALSE	FALSE	FALSE	FALSE	FALSE

Task 4 — When was the downloaded file executed by the victim according to Amcache?

Investigation

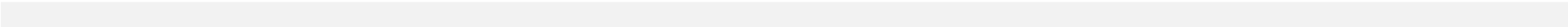
Amcache.hve (C:\Windows\appcompat\Programs\Amcache.hve) is a Windows registry hive that records program execution metadata — critically, the first time a binary was run on the system. I loaded Amcache.hve into Registry Explorer and searched for premium.exe.

The forensic entry showed:

```
Root\InventoryApplicationFile\premium.exe|45f3dc9b442b7d80
Last write timestamp: 2026-03-04 16:44:33
```

Registry Explorer v2.1.0
File Tools Options Bookmarks (0/0) View Help
Registry hives (2) Available bookmarks (0/0)
premium.ex Find
Key name # values # subkeys Last write timestamp
C:\Users\HP\Downloads\Fruitzy\A... 2025-10-27 18:26:44
{11517B7C-E79D-4e20-961B-75A81171... 2 1 2025-10-27 19:25:05
Root 0 26 2025-10-27 19:25:05
InventoryApplicationFile 2 476 2026-03-04 17:18:59
premium.exe|45f3dc9b442b7d80 9 0 2026-03-04 16:44:33
H:\C\Windows\AppCompat\Progra... 2025-10-27 18:26:44
{11517B7C-E79D-4e20-961B-75A81171... 2 1 2025-10-27 19:25:05
Root 0 26 2025-10-27 19:25:05
InventoryApplicationFile 2 476 2026-03-04 17:18:59
premium.exe|45f3dc9b44... 9 0 2026-03-04 16:44:33

Values
Drag a column header here to group by that column
Value Name Value Type Data Value Slack Is ... Data I
ProgramId RegSz 0006bab052e41661822b7b4bc649dcc5d59d0000ffff 02-00
FileId RegSz 0000405481d3d5529445c546d4429b72c5827ebde840 00-00
LowerCaseLongPath RegSz c:\users\cyberjunkie\downloads\premium.exe 00-00-01-00-0...
Name RegSz premium.exe 61-00-6D-00
BinaryType RegSz pe32_j386
LinkDate RegSz 04/27/2016 01:27:51 32-00-65-00
Size RegQword 11056040 38-29-06-00
Language RegDword 0
Usn RegQword 1369003752 80-3B-17-00



Task 5 — What is the SHA256 hash of the malicious executable?

Investigation

From Amcache metadata of premium.exe, I found the 'FileId' which is equivalent to SHA-1 hash (**SHA-1**: 405481d3d5529445c546d4425b72c5827ebde840-ignore first 4 digits). Then queried VirusTotal with the SHA-1 hash and got SHA-256 hash.

	Value Name	Value Type	Data	Value Slack	Is ...	Data Record Rea...
	ProgramId	RegSz	0006bab052e41661822b7b4bc649dcc5d59d0000ffff	02-00	☐	☐
	FileId	RegSz	0000405481d3d5529445c546d4429b72c5827ebde840	00-00	☐	☐
	LowerCaseLongPath	RegSz	c:\users\cyberjunkie\downloads\premium.exe	00-00-01-00-0...	☐	☐
	Name	RegSz	premium.exe	61-00-6D-00	☐	☐
	BinaryType	RegSz	pe32_i386		☐	☐
	LinkDate	RegSz	04/27/2016 01:27:51	32-00-65-00	☐	☐
	Size	RegQword	11056040	38-29-06-00	☐	☐
	Language	RegDword	0		☐	☐
	Usn	RegQword	1369003752	80-3B-17-00	☐	☐

SHA-256 Hash from VirusTotal : af240a2c2a4b42e3a130f47ccaab8aa2e20a1a588bc959ee9efd7475055ea7e3

Basic properties ⓘ	
MD5	b7f8099a29195303b8dbedec07a28bd0
SHA-1	405481d3d5529445c546d4429b72c5827ebde840
SHA-256	af240a2c2a4b42e3a130f47ccaab9aa2e20a1a586bc959ee9efd7475055ea7e3
Vhash	0170766d157c0d5d0d60a043z8003a7z47z62z3efz

Additional VirusTotal detail confirms that premiem.exes’s digital certificate also revoked by its issuer.

Signature info ⓘ	
Signature Verification	
⚠ A certificate was explicitly revoked by its issuer.	
File Version Information	
Date signed	2026-02-25 15:19:00 UTC

Task 6 — When was the Microsoft Defender scan of the file initiated?

Investigation

After executing the file and receiving no party invitation, the victim grew suspicious and manually initiated a Microsoft Defender custom scan on the file. To pinpoint the exact time, I opened Windows Event Viewer and loaded the saved log:

```
Microsoft-Windows-Windows Defender/Operational
```

Filtering for **Event ID 1000 (Defender scan started)** returned 5 matches. The most recent entry was dated 2026-03-04 and the event confirmed a Custom Scan targeting C:\Users\cyberjunkie\Downloads\premium.exe, initiated by user QA-PIPELINE-02\cyberjunkie on computer QA-Pipeline-02.

GeneralDetails

Microsoft Defender Antivirus scan has started.
Scan ID: {4814F09C-EF0D-4DAC-BBC8-99C081BAB913}
Scan Type: Antimalware
Scan Parameters: Custom Scan
Scan Resources: file: C:\Users\cyberjunkie\Downloads\premium.exe
User: QA-PIPELINE-02\cyberjunkie
Scan Trigger: Not specified
Scan Only If Idle: Enabled
Low CPU Priority for Scans: Disabled
Thread Priority: 7

Log Name:Microsoft-Windows-Windows Defender/Operational

Source:Windows Defender

Logged:3/4/2026 10:18:00 PM

Event ID:1000

Task Category:None

Level:Information

Keywords:

User:SYSTEM

Computer:QA-Pipeline-02

Save Filtered Log File As...

Save Filter to Custom View...

View

Delete

Rename

Refresh

Help

Event 1000, Windows Defender

Event Properties

Copy

Save Selected Events...

Refresh

Help

Then Switching to the **Details tab** > **Friendly View** and reading the **SystemTime** revealed the precise UTC timestamp:

2026-03-04T16:48:00.9828308Z

General

Details

☒ Friendly View ☐ XML View

- System

- Provider

[Name]

Microsoft-Windows-Windows Defender

[Guid]

{11cd958a-c507-4ef3-b3f2-5fd9dfbd2c78}

EventID

1000

Version

0

Level

4

Task

0

Opcode

0

Keywords

0x8000000000000000

- TimeCreated

[SystemTime]

2026-03-04T16:48:00.9828308Z

Task 7 — What was the RMM backdoor service name?

Investigation

To identify the installed backdoor, I filtered the Windows System event log for **Event ID 7045, new service was installed in the system.** A suspicious entry was logged at **2026-03-04 16:44:45 UTC** with the service name **CentraStage**. Its silent installation, configured with an *auto-start* trigger and *LocalSystem* privileges, is a clear indicator of a malicious backdoor deployment

```
Service Name:      CentraStage
Service File Name:  C:\Program Files (x86)\CentraStage\CagService.exe
Service Type:       user mode service
Service Start Type: auto start
Service Account:    LocalSystem
```

I dug deeper into **CentraStage**, which is the legitimate enterprise IT, it functions as a Remote Monitoring and Management (RMM) tool designed to let IT administrators manage networks, push patches, and execute commands remotely. Because it features powerful, built-in system management capabilities, threat actors frequently abuse it as a stealthy backdoor.

Filtered: Log: file://H:\C\Windows\System32\winevt\logs\System.evtx; Source: ; Event ID: 7045. Number of events: 23

Level	Date and Time	Source	Event ID	Task Category
Information	3/4/2026 10:14:45 PM	Service Control Manager	7045	None
Information	2/22/2026 11:54:29 PM	Service Control Manager	7045	None
Information	2/22/2026 11:54:28 PM	Service Control Manager	7045	None
Information	2/22/2026 10:32:59 PM	Service Control Manager	7045	None
Information	10/27/2025 3:03:20 PM	Service Control Manager	7045	None
Information	10/27/2025 3:03:16 PM	Service Control Manager	7045	None

Event 7045, Service Control Manager

GeneralDetails

A service was installed in the system.

Service Name: CentraStage
Service File Name: "C:\Program Files (x86)\CentraStage\CagService.exe"
Service Type: user mode service
Service Start Type: auto start
Service Account: LocalSystem

System_3

Open Saved Log...

Create Custom View...

Import Custom View...

Filter Current Log...

Clear Filter

Properties

Find...

Save Filtered Log File As...

Save Filter to Custom View...

View

Delete

Rename

Refresh

Help

Task 8 — What was the timestamped modified timestamp set on the RMM executables?

Investigation

- **Timestomping** is an anti-forensics technique where an attacker alters a file's timestamps to make it look older, helping it evade timeline-based detection. To investigate this, I used **MFTECmd** to parse the Master File Table (MFT), exported the results to a CSV file, and filtered the ParentPath column for CentraStage files. The file Gui.exe (located in .\Program Files (x86)\CentraStage) showed clear timestamp mismatches across two critical MFT attributes:
- **\$STANDARD_INFORMATION (\$SI / 0x10 attributes):** These handle standard creation and modification times (Created0x10 and LastModified0x10). They can be modified via normal Windows API calls, making them vulnerable to attacker manipulation.
- **\$FILE_NAME (\$FN / 0x30 attributes):** These record the kernel-level creation and rename timestamps (Created0x30 and LastModified0x30). They are written directly by the OS kernel and cannot be altered by standard user-mode API calls.
- The divergence between the manipulated \$SI timestamps (2/9/2026 07:56:40 UTC) and the ground-truth \$FN kernel timestamps (3/4/2026 16:44:45 UTC) provides definitive proof of timestomping.

T226119	:	✕	✓	f _x	2/9/2026 7:56:40 AM													
	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W
					Refer	Rep												
					ence	arse	IsDir	Ha										
					FileSi	Coun	Targ	ector	sA		uSecZero			NameTyp				
1	ParentPath	FileName	Extensi	ze	t	e	y	c	IsAds	SI<FN	s	Copied	SiFlags	e	Created0x10	Created0x30	LastModified0x	LastModified0x30
140582	.\Windows\Temp\forensicscopy\																	
	C:\Windows\prefetch	GUI.EXE-DED2786C.pf	.pf	####	1	####	###	FALSE	TRUE	FALSE	FALSE	Archive	Windows		3/4/2026 16:44	3/4/2026 17:20	3/4/2026 16:44	3/4/2026 17:20
226119	.\Program Files (x86)\CentraStage	Gui.exe	.exe	####	1	####	###	FALSE	TRUE	TRUE	FALSE	Archive	DosWindc		2/9/2026 7:56	3/4/2026 16:44	2/9/2026 7:56	3/4/2026 16:44

Task 9 — What is the name of the company whose product is the RMM tool?

Investigation

CentraStage is the legacy/agent name for Datto RMM, a product of Datto, Inc. CentraStage is widely used by managed service providers for legitimate remote IT management — which is precisely why threat actors abuse it. Its signed binaries and legitimate network traffic blend seamlessly with normal business activity, making it difficult to detect and rarely blocked by firewalls.

Task 10 — When was the malicious domain registered?

Investigation

PhishTool’s URL analysis exposed the domain. Then I queried ‘pomi.digital’ on VirusTotal and under registration data, got the following info:

```
Registration: 2026-02-20T01:06:05Z
Expiration: 2027-02-20T01:06:05Z
```

The domain was registered just 12 days before the phishing email was sent (2026-03-04). This is a classic threat actor pattern — freshly registered domains used for short-lived phishing campaigns before being burned and abandoned.

Details Authentication URLs Attachments Transmission X-headers

Filters (0) ▼

URL	 https://pomi.digital/premium?__cf_chl_tk=6SX3C1utT5mfbwmfIZyMDU.WICiyPkGsBkaOTitVBss-1772039836-1.0.1.1-JvMwxVsDmAF0zIVuaF0XNh73a8usyjzPeFDOJpaT4j8	...
Domain	 pomi.digital	

 pomi.digital

Registration Data (RDAP) ⓘ

Name:

pomi.digital

Handle:

dfc0f083657a4814a9be1b136674abd1-DONUTS

Events:

registration:

2026-02-20T01:06:05Z

expiration:

2027-02-20T01:06:05Z

last changed:

2026-04-21T01:19:07Z

last update of RDAP database:

2026-04-21T01:19:07Z

Task 11 — What is another name for the initial executable?

Investigation

In VirusTotal's Details tab for the premium.exe hash, the Names section lists all filenames under which this binary has been submitted:

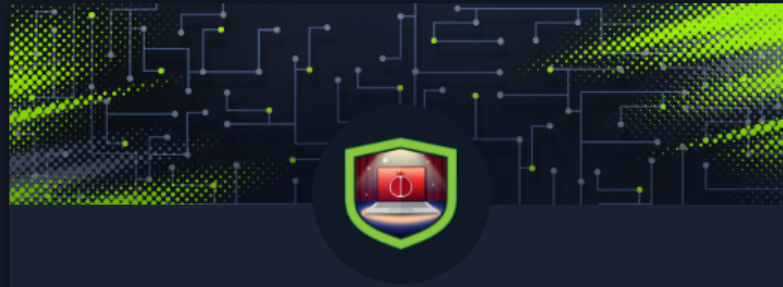
premium.exe

5bxx.exe

Names ⓘ

premium.exe

5bxx.exe



Fruity has been Solved

RismaFareedh has successfully solved Fruity from Hack The Box

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SHERLOCK RANK

01 May 2026

Powered by  **HACKTHEBOX**

SOLVE DATE

<https://labs.hackthebox.com/achievement/sherlock/2413013/1212>